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Section: F
Assignment: 03

1. $x^2 + y^2 = r^2$

- $0 \leq r^2 \leq a^2$
- $0 \leq r \leq a$

$$I_x = \sigma \int_0^{2\pi}\int_0^{a} r^2 \sin^2\theta \, r \, dr d\theta$$

$$= \sigma \int_0^{2\pi} \frac{r^4 \sin^2\theta}{4}\Big|_0^a d\theta$$

$$= \sigma \int_0^{2\pi} \frac{a^4 \sin^2\theta}{4} d\theta$$

$$= \sigma \int_0^{2\pi} \frac{a^4 (1-\cos 2\theta)}{8} d\theta$$

$$= \frac{\sigma}{8}\left[a^4\theta - \frac{a^4 \sin 2\theta}{2}\Big|_0^{2\pi}\right]$$

$$I_y = \sigma \int_0^{2\pi}\int_0^{a} r^2 \cos^2\theta \, r dr d\theta$$

$$= \sigma \int_0^{2\pi}\int_0^{a} r^3 \cos^2\theta \, dr d\theta$$

$$= \sigma \int_0^{2\pi} \frac{r^4 \cos^2\theta}{4}\Big|_0^a d\theta$$

$$= \sigma \int_0^{2\pi} \frac{a^4 \cos^2\theta}{4} d\theta$$

$$= \sigma \int_0^{2\pi} \frac{a^4 (1+\cos 2\theta)}{8} d\theta$$

$$= \frac{\sigma}{8}\int_0^{2\pi} a^4 + a^4 \cos 2\theta \, d\theta$$

$$= \frac{\sigma}{8}\left[a^4\theta + \frac{a^4 \sin 2\theta}{2}\Big|_0^{2\pi}\right]$$

$$= \frac{\sigma}{8}(2a^4\pi)$$

$$\frac{\sigma\pi a^4}{4} = I_y = I_x$$

$$I_z = I_y + I_x$$

$$\frac{\sigma\pi a^4}{4} + \frac{\sigma\pi a^4}{4}$$

$$I_z = \frac{\sigma\pi a^4}{2}$$

CamScanner

2. Avg Vol $= \frac{1}{V} \cdot \iiint f(x,y,z)\, dV$

Volume $= \int_0^1 \int_0^{1-x} \int_0^1 1 \cdot dz\, dy\, dx$

$= \int_0^1 \int_0^{1-x} z \Big|_0^1 \, dy\, dx$

$= \int_0^1 y \Big|_0^{1-x} \, dx$

$= \int_0^1 1-x \, dx$

$= x - \frac{x^2}{2} \Big|_0^1$

Volume $= \frac{1}{2}$

Average $= 2 \times \int_0^1 \int_0^{1-x} \int_0^1 x+y+z \; dz\, dy\, dx$

$= \int_0^1 \int_0^1 xz + yz + \frac{z^2}{2} \Big|_0^1 \; dy\, dx$

$= \int_0^1 \int_0^1 x + y + \frac{1}{2} \; dy\, dx$

$= \int_0^1 xy + \frac{y^2}{2} + \frac{1y}{2} \Big|_0^1 \; dx$

$= \int_0^1 x + 1 \; dx$

$= \frac{x^2}{2} + x \Big|_0^1$

Average $= 3$

3, $x = \cos t$, $y = \sin t$, $z = t/4$ $\quad 0 \le t \le 4\pi$

Length of helical axis:

$$L = \int ds$$

$$= \int_0^{4\pi} \sqrt{dx^2 + dy^2 + dz^2}$$

$$= \int_0^{4\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1/4)^2}\, dt$$

$$= \int_0^{4\pi} \sqrt{1 + 1/16}\, dt$$

$$L = \sqrt{17/16} \int_0^{4\pi} dt = \frac{\sqrt{17}}{4} 4\pi = \sqrt{17}\,\pi$$

the cross section is a circle with radius $1/2$,

$$A = \pi r^2 = \pi (1/2)^2 = \frac{1}{4}\pi$$

$$V = L \cdot A$$

$$= \sqrt{17}\,\pi \cdot \frac{\pi}{4}$$

$$V = \frac{\pi^2 \sqrt{17}}{4}$$

4, $z = \dfrac{1}{x^2 + y^2 + 1}$, $z = \dfrac{1}{r^2 + 1}$

a, $r = a$, $z = \dfrac{1}{a^2 + 1}$

As $a \to 0^+$, z approaches to 1

As a approaches $+\infty$, z approaches 0

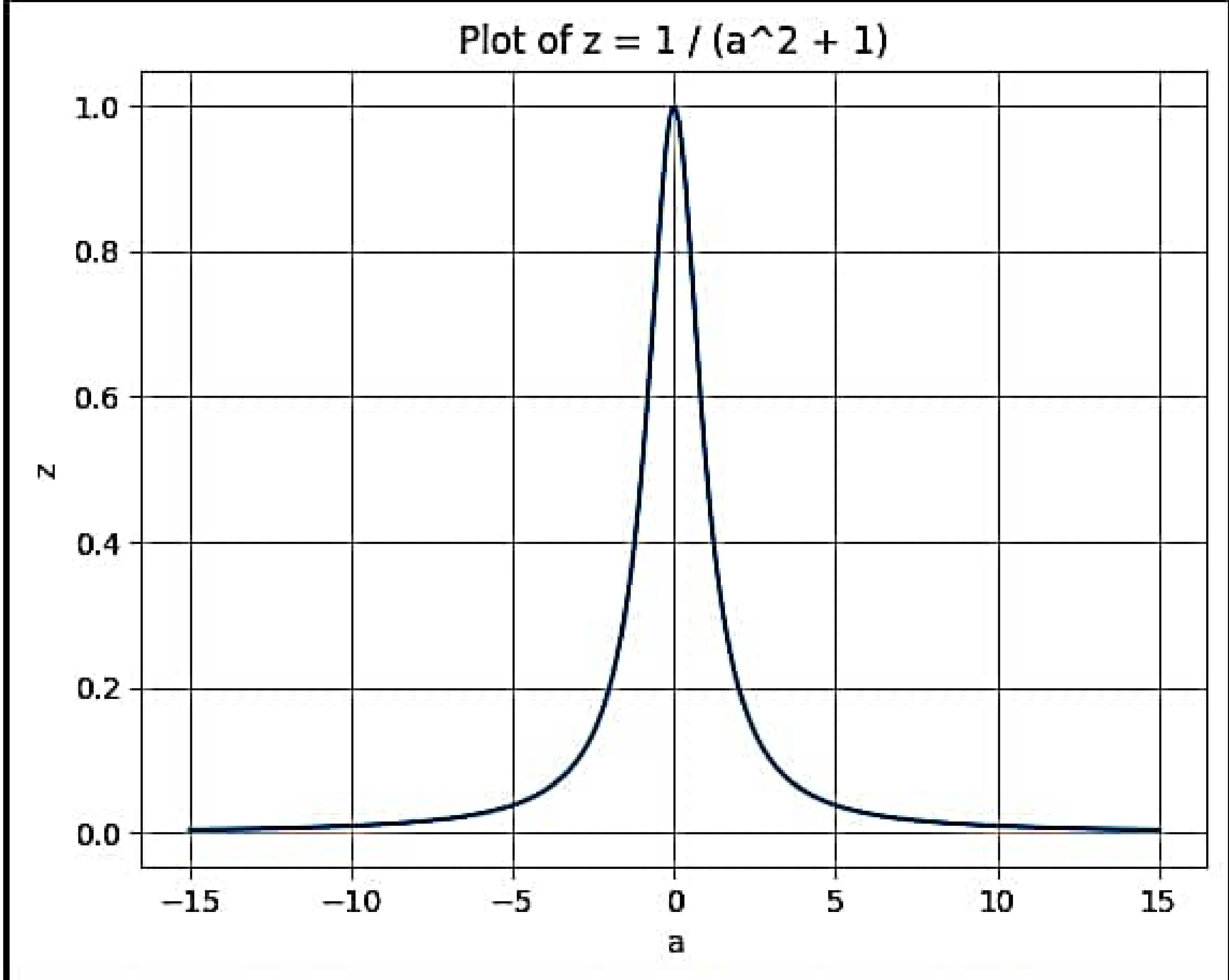
As y-axis is expanding, z is reducing to 0

b, $\lim\limits_{a \to 0^-} z = \dfrac{1}{1} = 1$

$\lim\limits_{a \to \infty} = \dfrac{1}{\infty} = 0$

c, python

```python
import numpy as np
import matplotlib.pyplot as plt

a_values = np.linspace(-15, 15, 300)

z_values = 1 / (a_values ** 2 + 1)

plt.plot(a_values, z_values)
plt.xlabel('a')
plt.ylabel('z')
plt.title('Plot of z = 1 / (a^2 + 1)')
plt.grid(True)
plt.show()

z_target = 0.25
a_at_z_target = np.sqrt(1 / z_target - 1)

print("The value of a for which the z-coordinate is 0.25 is:", a_at_z_target)
```




```
The value of a for which the z-coordinate is 0.25 is: 1.7320508075688772
```

5. $\frac{2}{3}x = 2\sqrt{x}$

$\frac{x}{3} = \sqrt{x}$

$\frac{x^2}{9} - x = 0$

$= \int_0^9 \int_{\frac{2x}{3}}^{2\sqrt{x}} 2yx^2 + 9y^3 \, dy \, dx$

$= \int_0^9 y^2x^2 + \frac{9y^4}{4} \Big|_{\frac{2x}{3}}^{2\sqrt{x}} dx$

$= \int_0^9 4x(x^2) + \frac{9(16x^2)}{4} \, dx$

$= \int_0^9 4x^3 + 36x^2 \, dx$

$= x^4 + 12x^3 \Big|_0^9 = 15309$

6. Python

7. $x^2+y^2 = 2y$

$r^2 = 2r\sin\theta \qquad 0 \le \theta \le \pi$

$r = 2\sin\theta$

$\int_0^{\pi} \int_0^{2\sin\theta} r r \, dr \, d\theta$

$\int_0^{\pi} \frac{1}{3} r^3 \Big|_0^{2\sin\theta} d\theta$

$\frac{8}{3} \int_0^{\pi} \sin^3\theta \, d\theta$

$\frac{8}{3} \int \sin\theta - \frac{8}{3} \int \sin\theta \cos^2\theta \, d\theta$

$\frac{8}{3} \left[-\cos\theta \Big|_0^{\pi} + \frac{\cos^3\theta}{3} \Big|_0^{\pi} \right]$

$\frac{8}{3}\left(2 - \frac{2}{3}\right)$

$\frac{32}{9}$

$\sin^3\theta = \sin^2\theta \sin\theta$

$= \sin\theta(1 - \cos^2\theta)$

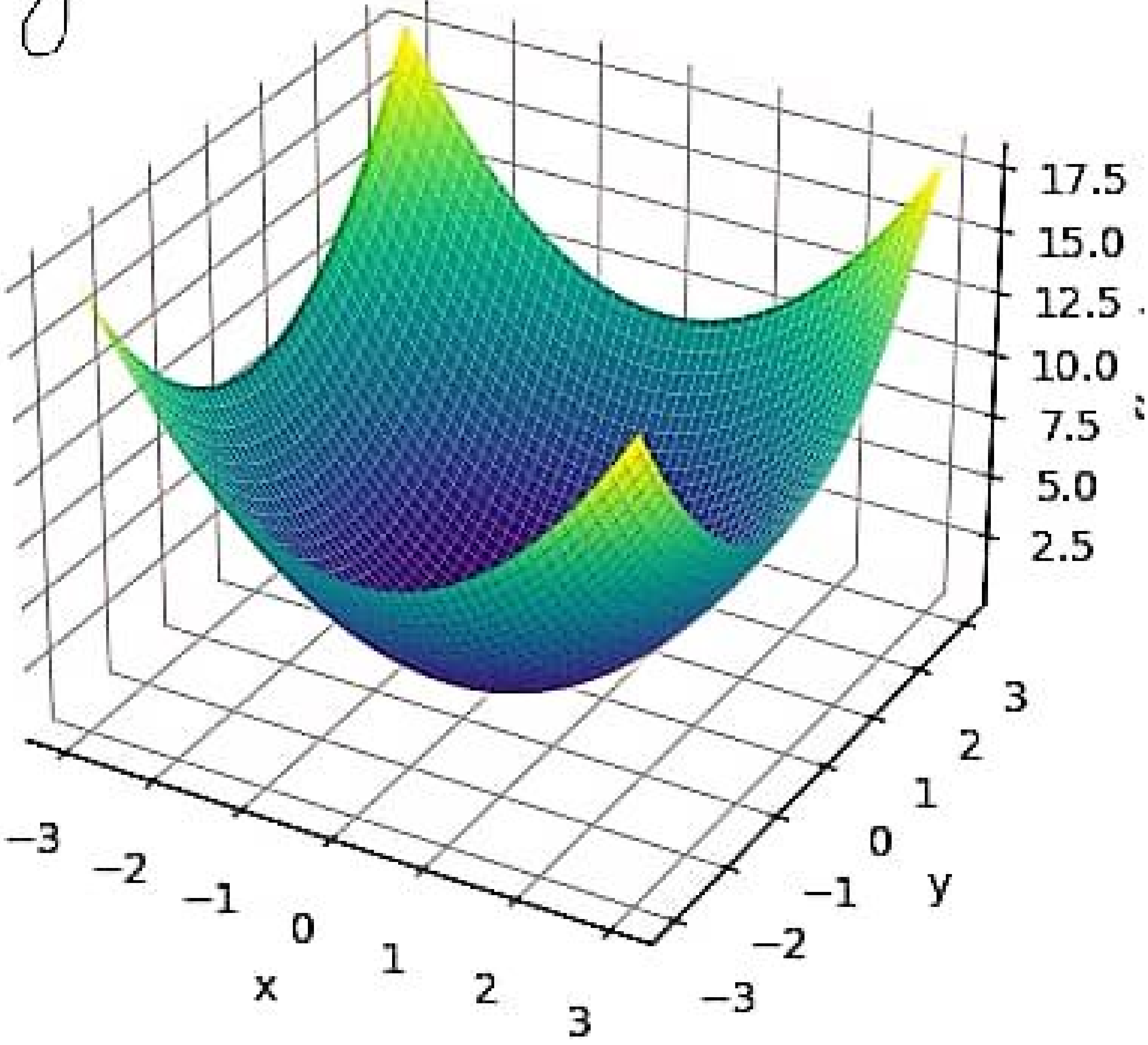
$\sin^3\theta = \sin\theta - (\sin\theta)(\cos^2\theta)$

8-10 - Python

Q6

```python
def integrand(x, y, z):
    return x**2 + y**2 + z**2

def triple_integral():
    integral_result = 0
    num_steps = 100

    dx = 6 / num_steps
    dy = 4 / num_steps
    dz = 3 / num_steps

    for i in range(num_steps):
        x = i * dx
        for j in range(num_steps):
            y = j * dy
            for k in range(num_steps):
                z = k * dz
                if 2*x + 3*y + 4*z <= 12:
                    integral_result += integrand(x, y, z) * dx * dy * dz

    return integral_result

result = triple_integral()
print("Result of the triple integral:", result)
```

```
Result of the triple integral: 78.69220582319998
```


17.5
15.0
12.5
10.0
7.5
5.0
2.5
−3
−2
−1
0
1
2
3
x
3
2
1
0
−1
−2
−3
y

```
import numpy as np
import matplotlib.pyplot as plt

def f(x, y):
  return x**2 + y**2

x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)
Z = f(X, Y)

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, Z, cmap='viridis')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('f(x, y)')
plt.show()
```

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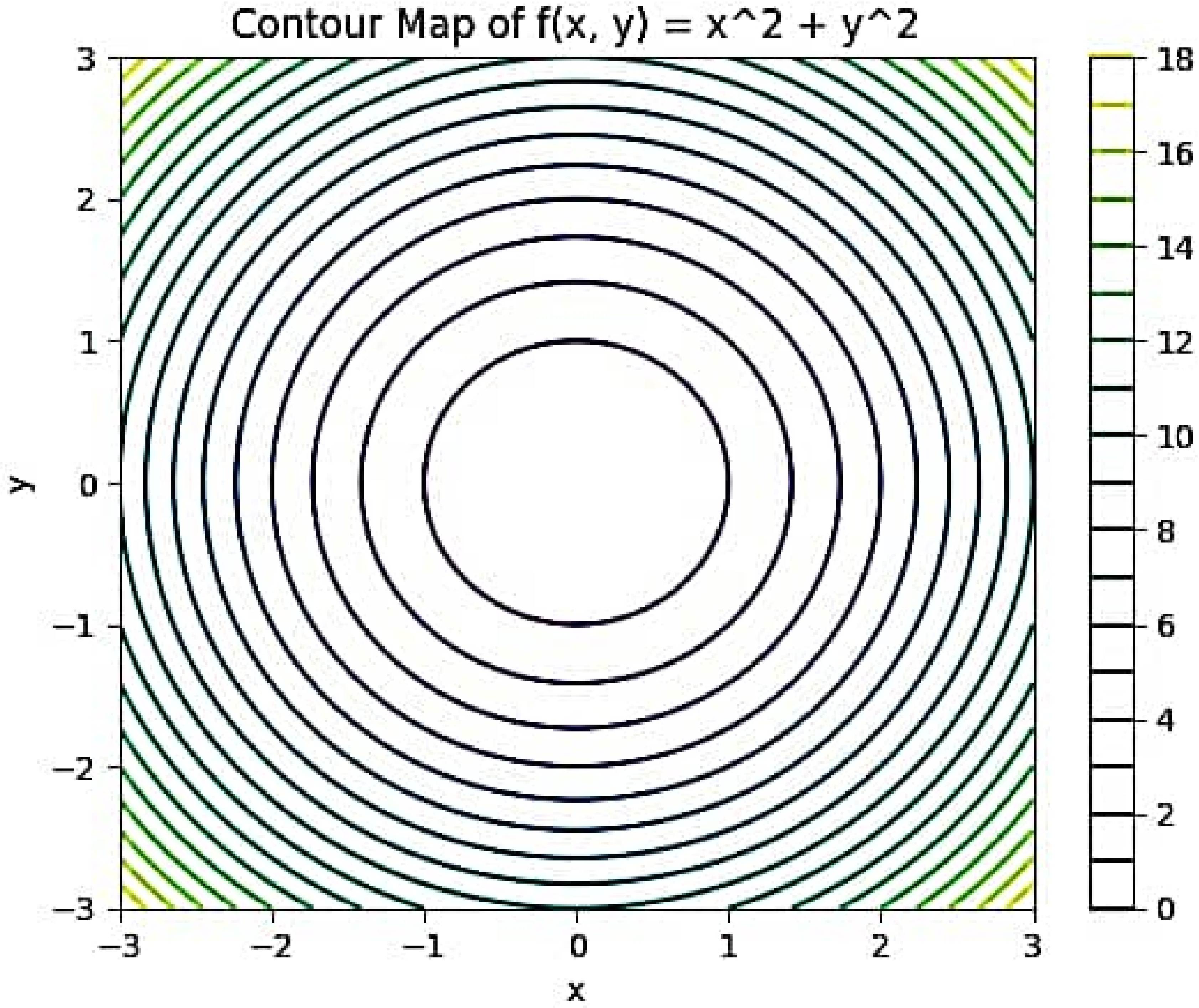
```
import numpy as np
import matplotlib.pyplot as plt

def f(x, y):
  return x**2 + y**2

x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)

Z = f(X, Y)

plt.contour(X, Y, Z, levels=20)
plt.colorbar()
plt.title('Contour Map of f(x, y) = x^2 + y^2')
plt.xlabel('x')
plt.ylabel('y')
plt.show()
```



Contour Map of f(x, y) = x^2 + y^2
x
y
-3
-2
-1
0
1
2
3
0
2
4
6
8
10
12
14
16
18

```python
import numpy as np
import matplotlib.pyplot as plt

def f(x, y):
  return x*2 + y*2

x = np.linspace(-3, 3, 50)
y = np.linspace(-3, 3, 50)
X, Y = np.meshgrid(x, y)

gx = 2 * X
gy = 2 * Y

plt.quiver(X, Y, gx, gy)
plt.title('Gradient Vector Field of f(x, y) = x^2 + y^2')
plt.show()
```

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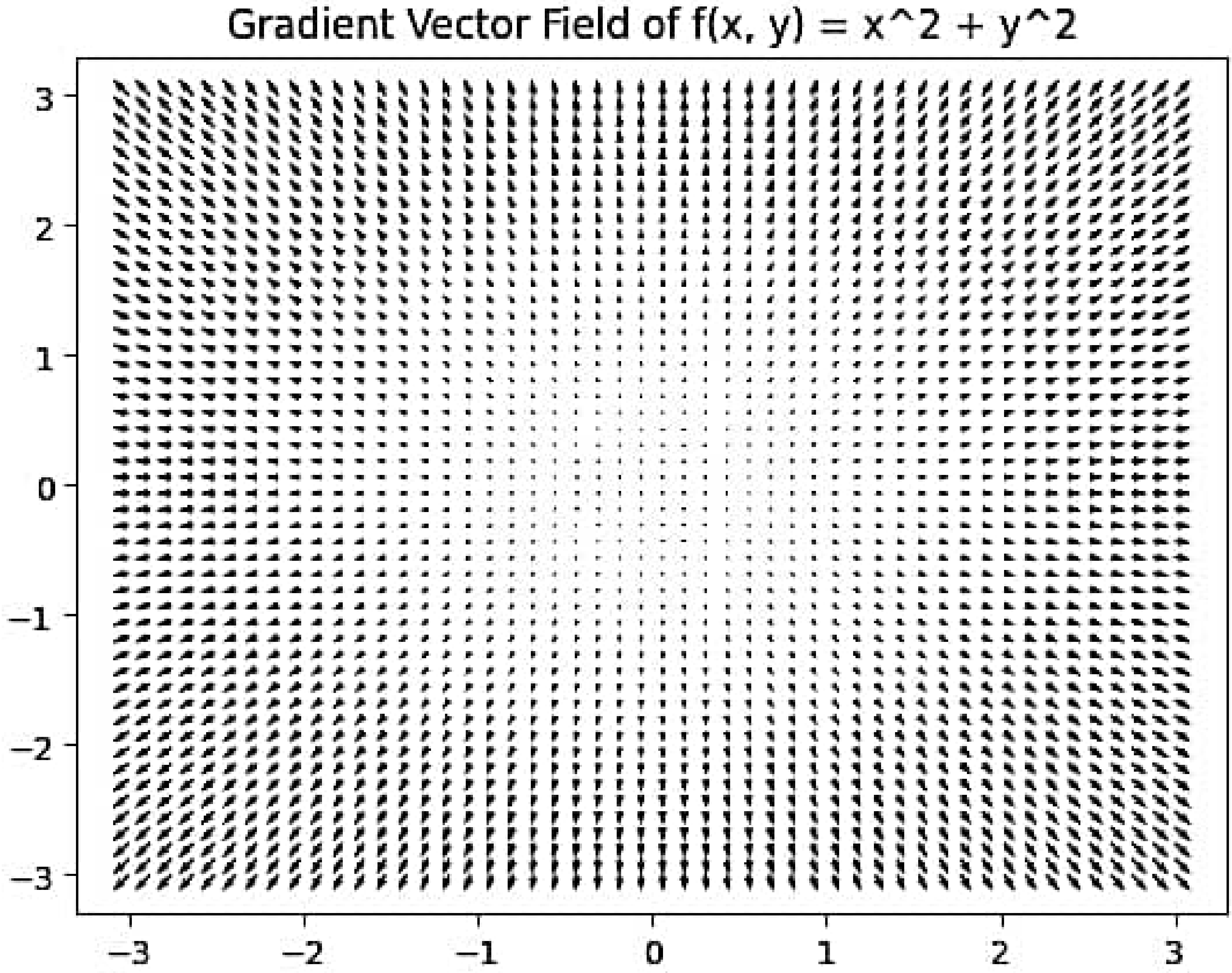

Gradient Vector Field of f(x, y) = x^2 + y^2
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3

11. $I_0 = (4, 2, -1)$ $\qquad f(x_1, x_2, x_3) = (x_1 - 4)^4 + (x_2 - 3)^2 + 4(x_3 + 5)^4$

$I_1 = I_0 - \lambda_0 (\nabla f(I_0))$ $\qquad \nabla f = 4(x_1 - 4)^3 \hat{i} + 2(x_2 + 3)\hat{j} + 16(x_3 + 5)^3 \hat{k}$

$f(I_1) = (4-4)^4 + (2 - 2\lambda_0 - 3)^2 + 4(-1 - 1024\lambda_0 + 5)^4$ $\qquad \nabla f = 0\hat{i} + 2\hat{j} + 0\hat{k}$

$= (-2\lambda_0 - 1)^2 + 4(4 - 1024\lambda_0)^4$ $\qquad \nabla f(I_1) = 0\hat{i} + 2\hat{j} + 0\hat{k}$

$\nabla f(I_2) = (0, 0, 0)$

$f'(I_1) = 2(-2\lambda_0 - 1)(-2) + 16(4 - 1024\lambda_0)(t - 1024)$

$0 = 8\lambda_0 + 4 - 65536 + 16777216\lambda_0$

$65532 = 16777224\lambda_0$

$\lambda_0 = 3.9 \times 10^{-3}$

$I_2 = I_1 - \lambda_1 (\nabla f(I_1))$

$I_2 = \begin{pmatrix} 4 \\ 2 \\ 5 \end{pmatrix} - \lambda_1 \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = (2^4 - 2\lambda - 5)$

$f(I_2) = 0 + (2 - 2\lambda_2 - 3)^2 + 4(0)^4$

$= (-1 - 2\lambda_2)^2$

$f'(I_2) = 2(-1 - 2\lambda_2)(-2)$

$0 = 4 + 8\lambda_2$

$\lambda_2 = -1/2$ $\qquad I_2 = \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix}$

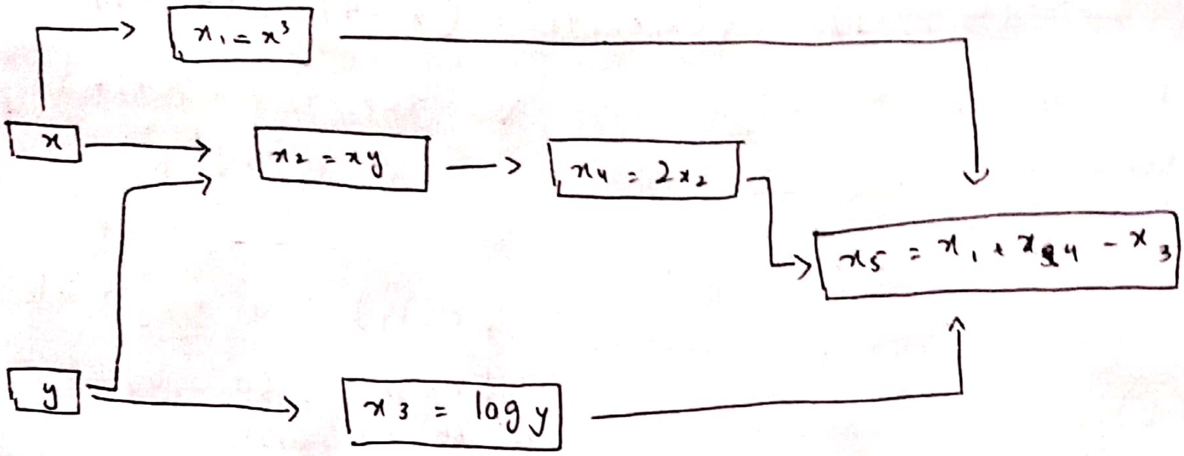
$I_3 = I_2 - \lambda_2 (\nabla f(I_2))$

$= (4, 3, -5) - \lambda_2 (0, 0, 0) = \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix}$

$I_0 = (4, 2, -1)$ $\qquad I_1 = (4, 2, -5)$

$I_2 = (4, 3; 5)$ $\qquad I_3 = (4, 3, -5)$

12.



$$\frac{\partial x_3}{\partial y} = \frac{1}{\ln 10}, \quad \frac{\partial x_1}{\partial x} = 8, \quad \frac{\partial x_2}{\partial y} = 2, \quad \frac{\partial x^4}{\partial x_2} = 2$$

$$\frac{\partial x^5}{\partial x^4} = 1, \quad \frac{\partial x^5}{\partial x_4} = 1, \quad \frac{\partial x_5}{\partial x_3} = 1$$

$$\nabla f = \left((1\times 8) + (1\times 2x), \ (1\times 2x \times 2) + \left(-1 + \frac{1}{\ln 10}\right)\right)$$

$$= (10, \ 3.57)$$

$$\nabla f = 10\,\hat{i} + 3.57\,\hat{j}$$